

Name: Find Path

Inputs:

- **graph**: a Graph object with edges representing the connections between people
- **start_person**: a node in **graph** representing the starting node
- **end_person**: a node in **graph** representing the ending node
- **parents**: a map representing the parents in the graph

Outputs:

- **path**: a sequence of tuples of the path in the form [(actor1, {movie1a, ...}), (actor2, {movie2a, ...}), ...]
1. **path** $\leftarrow$ an empty sequence
 2. append the tuple (**end_person**, empty set) at the end of the sequence **path**
 3. **current_node** $\leftarrow$ **end_person**
 4. while **current_node** is not equal to **start_person** do
 - a. **parent_node** $\leftarrow$ **parents**_{**current_node**}
 - b. if **parent_node** is null then
 - i. return an empty sequence
 - c. **attrs** $\leftarrow$ the set of attributes on the edge (**parent_node**, **current_node**) in **graph**
 - d. append the tuple (**parent_node**, **attrs**)
 - e. **current_node** $\leftarrow$ **parent_node**
 5. **final_path** $\leftarrow$ an empty sequence
 6. for **idx** in 0,1, ..., , length of **path** - 1 do
 - a. append **path**_{**idx**} at the end of the sequence of **final_path**
 7. Return **final_path**

Discussion

1. The diameter of a graph is defined as the maximum distance between any two nodes in the graph. Describe a strategy using what you have done in this assignment for computing the diameter of a graph.

Run BFS on all pairs of nodes in the graph and record the distances found. The maximum distance among all these distances is the diameter of the graph.

2. In the Kevin Bacon Game, highly connected actors are "valuable". The most connected nodes — in this case actors — in a graph are called "hubs". Describe a strategy for computing the hubs in a graph.

Iterate over all the nodes in the graphs and record the number of connections/edges each of the nodes have. Record the max degree. Iterate through the graph and add any nodes that have the max degree to a set. All the nodes in the set are hubs.

3. Just as certain actors can be identified as hubs in a movie graph, and therefore be seen as important to the structure to the graph, certain movies can also be considered more important to the structure of the graph. What criteria would you use to classify movies as such?

There are two criteria. First, a movie is important if a movie is unique in connecting two actors, meaning there is no other movie in the attribute of the edge between the two actors. Second, if the movie appears in many shortest paths between two nodes of the graph.

4. The `run` function uses the `simpleplot` module to plot the output of `distance_histogram` on two actors, Kevin Bacon and Stephanie Fratus, in the 5000-node subgraph. Describe the differences between these two plots; what does this tell you about these two actors?

Kevin Bacon's histogram shows the distance frequency is mostly 2 and 3, with a few connections being 1 and 4. However, Stephanie Fratus' histogram shows her distance frequency is primarily 3 and 4, with a few actors being up to 6 movies away from her. Since the distance between Bacon and the other actors is on average less than Fratus, this tells us that Bacon is more central to the graph and acts like a hub, while Fratus is less influential and connected in the graph.